

Financial Analysis and Decision Making (FADM) Global BBA Fall 2026

Jason Kang

Session 1: Time Value of Money Part 2

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Grading and Key Dates

- ▶ In-Class Test (Oct 1, ~ 1 hour): 15%
- * Midterm Exam (Oct 12-16): 30%
- ▶ Portfolio in Practice: In-Class Exam (Nov 19, ~ 1 hour): 15%
- * Final Exam (Dec 3-11): 40%

If you miss **one** of the In-Class Test or the PiP exam, final exam = 55%

If you miss **both**, you will get 0%

Plan For Today

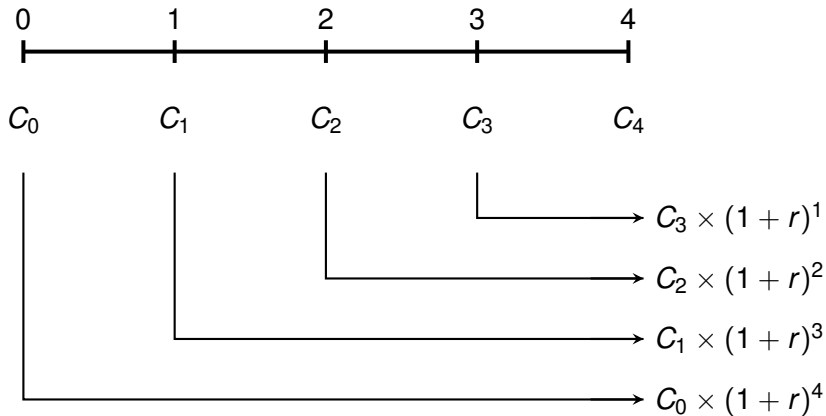
- ▶ Review
- ▶ Annuity
- ▶ More perpetuity/annuity

Suppose you are scheduled to receive the below stream of cash flows. Let $r = 10\%$. What is the value of the cash flows at the end of 3 years?



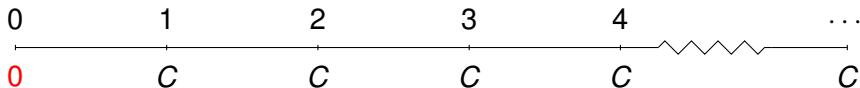
$$FV_3 = 5 \times 1.1^2 + 5 \times 1.1 + 105 = 116.55$$

The value of C in 4 years ($= FV_4$)



We can add these cash flows because they are in same (future) time unit

A regular perpetuity

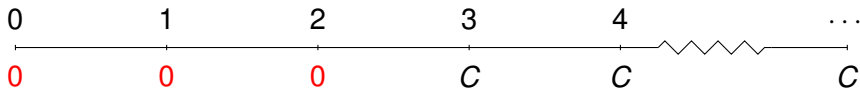


at time $t = 0$ has value

$$PV_0 = \frac{C}{r}$$

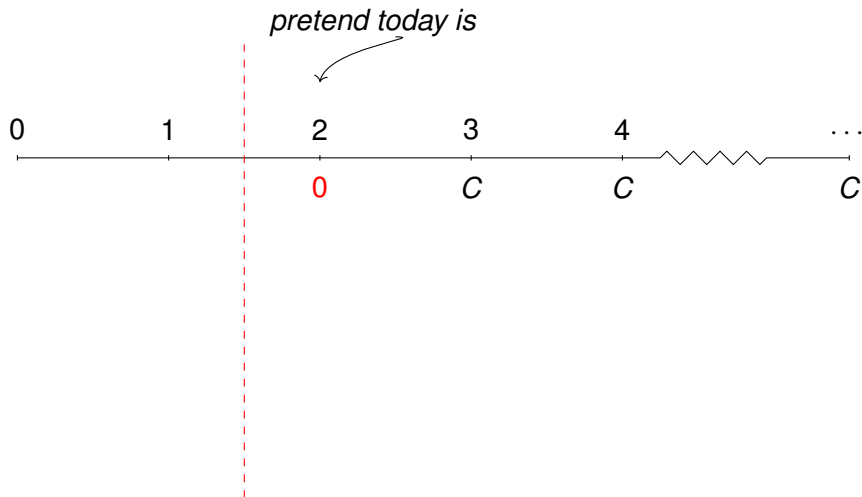
where PV_0 denotes present value in the perspective of time 0.

A deferred perpetuity

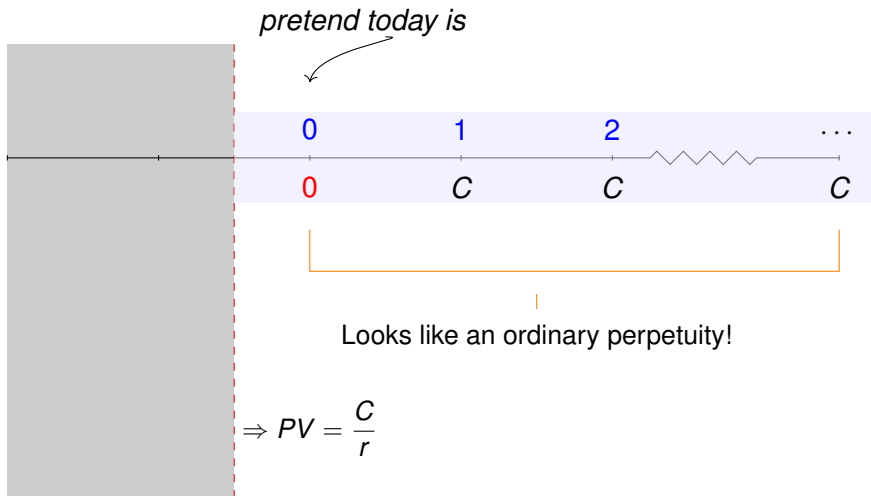


What if the first payment starts at $t = 3$?

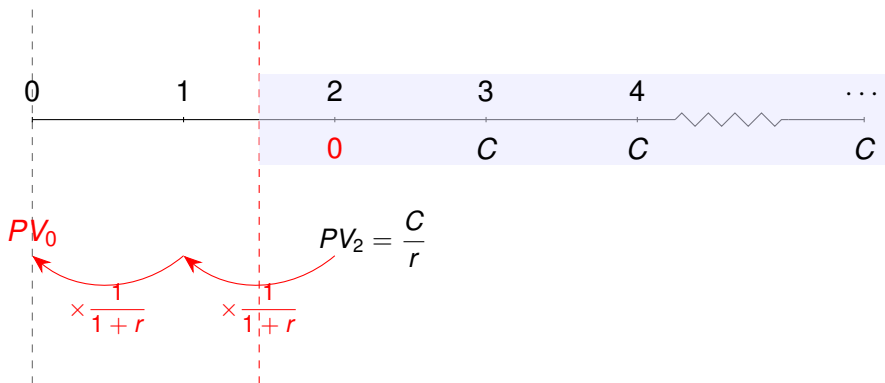
A deferred perpetuity



A deferred perpetuity



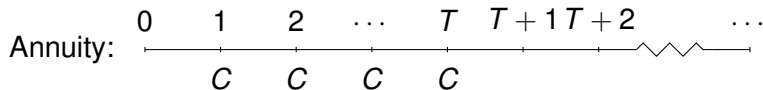
A deferred perpetuity



$$PV_0 = PV_2 \left(\frac{1}{1+r} \right)^2 = \frac{C}{r} \left(\frac{1}{1+r} \right)^2 = \frac{C}{r} \left(\frac{1}{1+r} \right)^{t-1}$$

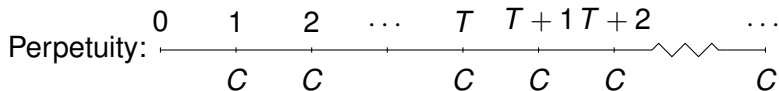
where t is the date of the first cash flow.

(Informal) Definition. You get a stream of cash flows T times.

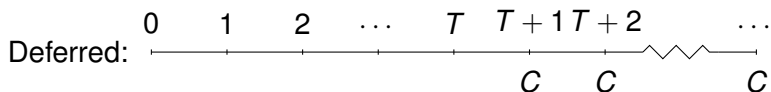


Like a perpetuity, the first payment occurs at $t = 1$.

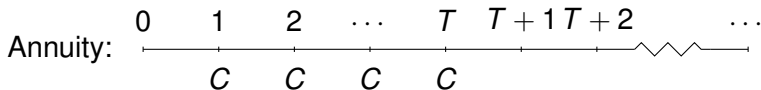
Annuity



—



=



Annuity = Perpetuity – Deferred Perp.

$$\begin{aligned} &= \frac{C}{r} - \frac{C}{r} \left(\frac{1}{1+r} \right)^{t-1} \\ &= \frac{C}{r} - \frac{C}{r} \left(\frac{1}{1+r} \right)^T \\ &= \frac{C}{r} \left[1 - \left(\frac{1}{1+r} \right)^T \right] \end{aligned}$$

T is the **number** of payments (remaining).

Example

(Slide 16 in FADM Session 1 part 2)

To purchase a house, you took out a 30-year mortgage with $r = 6\%$. The annual payment on the mortgage is \$12. **You just made a payment** and have decided to pay off the remaining balance. What is the payoff amount if:

- a. You have lived in the house for 12 years (so there are 18 years left)

$$\text{using } PV = \frac{C}{r} \left[1 - \left(\frac{1}{1+r} \right)^T \right]$$

where T is the number of payments remaining,

$$PV = \frac{12}{.06} \left[1 - \left(\frac{1}{1.06} \right)^{18} \right] = 129.9$$

Example

(Slide 16 in FADM Session 1 part 2)

To purchase a house, you took out a 30-year mortgage with $r = 6\%$. The annual payment on the mortgage is \$12.

- c. What is the outstanding balance **before** the 12th payment is made?

(Slide 16 in FADM Session 1 part 2)

To purchase a house, you took out a 30-year mortgage with $r = 6\%$. The annual payment on the mortgage is \$12.

- c. What is the outstanding balance **before** the 12th payment is made?

The general formula for an annuity looks like:

$$PV = \frac{C}{r} \left[1 - \left(\frac{1}{1+r} \right)^T \right] \left(\frac{1}{1+r} \right)^{t-1}$$

where

- ▶ T is the number of payments
- ▶ t is the date of the first payment

Example

(Slide 16 in FADM Session 1 part 2)

To purchase a house, you took out a 30-year mortgage with $r = 6\%$. The annual payment on the mortgage is \$12.

- c. What is the outstanding balance **before** the 12th payment is made?

Then,

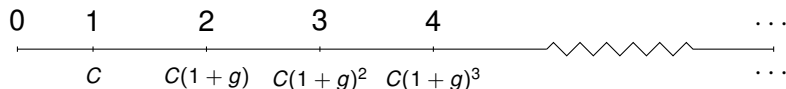
$$\begin{aligned}PV &= \frac{C}{r} \left[1 - \left(\frac{1}{1+r} \right)^T \right] \left(\frac{1}{1+r} \right)^{t-1} \\&= \frac{C}{r} \left[1 - \left(\frac{1}{1+r} \right)^{19} \right] \left(\frac{1}{1+r} \right)^{0-1} \\&= \frac{12}{.06} \left[1 - \left(\frac{1}{1.06} \right)^{19} \right] \times 1.06 = 141.9\end{aligned}$$

Coincidentally,

$$129.9 + 12 = 141.9$$

The hardest part...

Definition. Given a base payment C , subsequent payments increase by a constant rate g forever.



Example. Let $C = 10$ and $g = 5\%$

$t = 1$: Get $C = 10$

$t = 2$: Get $C(1 + g) = 10 \times 1.05 = 10.5$

$t = 3$: Get $C(1 + g)^2 = 10.5 \times 1.05 = 11.025$

The present value of a growing perpetuity is given by

$$PV = \frac{C}{r - g}$$

where r is the interest/discount rate and g is the growth rate of cash flows.

(Slide 24 in FADM Session 1 part 2)

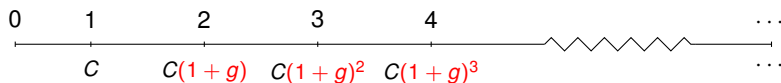
A rich relative has bequeathed you a growing perpetuity. The first payment will occur in a year and will be \$10. Each year after that, you will receive a payment on the anniversary of the last payment that is 8% larger than the last payment. This pattern of payments will go on forever. If the interest rate is 12% per year

- a . What is the value of the bequest today?

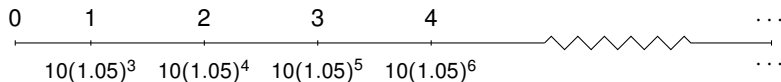
Trick 1

Trick 1

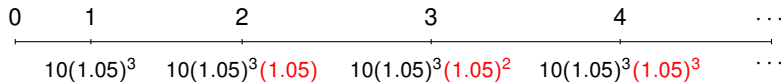
A growing perpetuity looks like this:



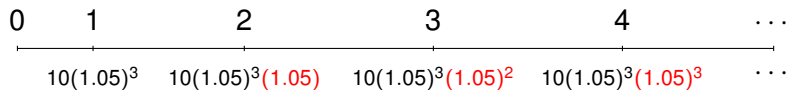
What if you have something that looks like this?



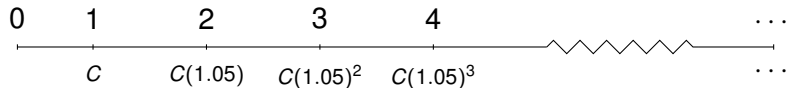
This is identical to:



Trick 1



Let $C = 10 \times 1.05^3$. Assume $r = 10\%$. Then,



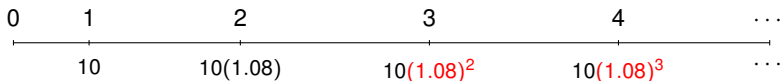
$$PV_0 = \frac{C}{r - g} = \frac{C}{.1 - .05} = \frac{10 \times 1.05^3}{.1 - .05}$$

Example

(Slide 24 in FADM Session 1 part 2)

A rich relative has bequeathed you a growing perpetuity. The first payment will occur in a year and will be \$10. Each year after that, you will receive a payment on the anniversary of the last payment that is 8% larger than the last payment. This pattern of payments will go on forever. If the interest rate is 12% per year

b . What is the value **immediately after** the first payment is made?



Example

(Slide 24 in FADM Session 1 part 2)

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- b . What is the value **immediately after** the first payment is made?

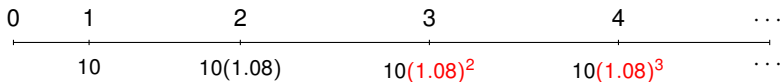
$$\begin{aligned}PV_1 &= \frac{C}{r - g} \\&= \frac{10 \times 1.08}{.12 - .08} \\&= \frac{10.8}{.04} = 270\end{aligned}$$

Example

(Slide 24 in FADM Session 1 part 2)

A rich relative has bequeathed you a growing perpetuity. The first payment will occur in a year and will be \$10. Each year after that, you will receive a payment on the anniversary of the last payment that is 8% larger than the last payment. This pattern of payments will go on forever. If the interest rate is 12% per year

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c . What is the value **immediately before** the first payment is made?

Using the formula for a deferred growing perpetuity,

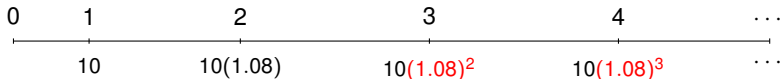
$$\begin{aligned} PV_1 &= \frac{C}{r-g} \left(\frac{1}{1+r} \right)^{t-1} \\ &= \frac{10}{.12 - .08} \left(\frac{1}{1.12} \right)^{0-1} \\ &= \frac{10}{.04} \times 1.12 = 280 \end{aligned}$$

Example

(Slide 24 in FADM Session 1 part 2)

A rich relative has bequeathed you a growing perpetuity. The first payment will occur in a year and will be \$10. Each year after that, you will receive a payment on the anniversary of the last payment that is 8% larger than the last payment. This pattern of payments will go on forever. If the interest rate is 12% per year

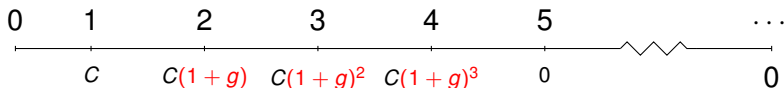
- c . What is the value **immediately before** the first payment is made?



Method 2: Let $C = 10(1.08)$. Then,

$$\begin{aligned} PV_1 &= 10 + \frac{C}{r - g} \\ &= 10 + \frac{10(1.08)}{.12 - .08} = 280 \end{aligned}$$

Growing Annuity



(Informal) Definition. You receive T payments. Your first payment, at $t = 1$, is C , then grows by rate g each year.

$$PV_0 = \frac{C}{r - g} \left[1 - \left(\frac{1 + g}{1 + r} \right)^T \right]$$

where T is the number of payments.

Example

Jason wants to sell trout at Skema. To catch trout, Jason needs to buy a boat today for €100. Starting next year, he expects to earn €10 in profits his first year, then growing at a rate of 2%. If Jason expects to work for 30 years, should he undertake the project? Because Jason is risky, assume the relevant discount rate is 15%.

$$\begin{aligned} NPV_0 &= -I + \frac{C}{r-g} \left[1 - \left(\frac{1+g}{1+r} \right)^T \right] \\ &= -100 + \frac{10}{.15 - .02} \times \left[1 - \left(\frac{1.02}{1.15} \right)^{30} \right] = -75, \text{ so NO.} \end{aligned}$$

(Slide 32 in FADM Session 1 part 2)

You're saving for retirement. You plan to contribute €12 to your account, starting the end of this year. You plan to increase your contributions by 3% each year for the next 45 years. Assume $r = 8\%$. How much will you have in your account when you retire in 45 years?

(Slide 32 in FADM Session 1 part 2)

You're saving for retirement. You plan to contribute €12 to your account, starting the end of this year. You plan to increase your contributions by 3% each year for the next 45 years. Assume $r = 8\%$. How much will you have in your account when you retire in 45 years?

$$\begin{aligned}PV_0 &= \frac{C}{r - g} \left[1 - \left(\frac{1 + g}{1 + r} \right)^T \right] \\&= \frac{12}{.08 - .03} \left[1 - \left(\frac{1.03}{1.08} \right)^{45} \right] = 211\end{aligned}$$

But we want FV_{45} ,

$$FV_{45} = PV_0(1 + r)^T = 211(1.08)^{45} = 6735$$

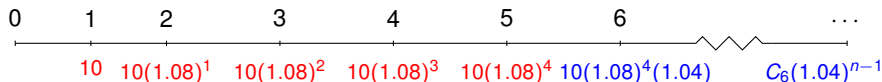
Trick 2

Trick 2

What is the value of a company's assets if it is expected to deliver cash flows starting next year at \$10m and growing at 8% for the following 4 years (up to year 5), and then at 4% in perpetuity? Assume a discount rate of 10%

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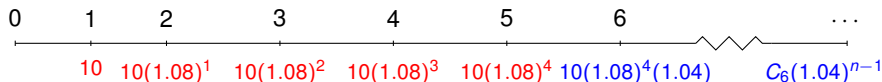


How can we decompose this asset?

$$PV_0 = \text{PV of growing annuity} + \text{PV of deferred, growing perpetuity}$$

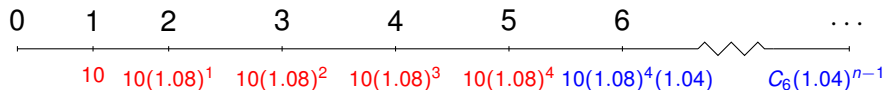
Trick 2

What is the value of a company's assets if it is expected to deliver cash flows starting next year at \$10m and growing at 8% for the following 4 years (up to year 5), and then at 4% in perpetuity? Assume a discount rate of 10%



$$\begin{aligned}PV_0 \text{ growing annuity} &= \frac{C}{r - g} \left[1 - \left(\frac{1 + g}{1 + r} \right)^T \right] \\&= \frac{10}{.02} \left[1 - \left(\frac{1.08}{1.1} \right)^5 \right] \\&= 43.83\end{aligned}$$

Trick 2

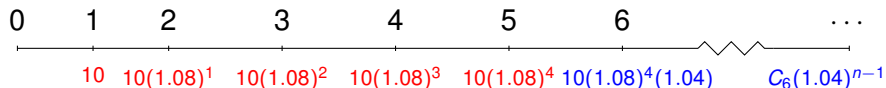


$$\begin{aligned}\text{PV of d,g perpetuity} &= \frac{C}{r-g} \left(\frac{1}{1+r} \right)^{t-1} \\ &= \frac{10 \times (1.08)^4 \times 1.04}{.10 - .04} \left(\frac{1}{1.1} \right)^5 \\ &= 146.41\end{aligned}$$

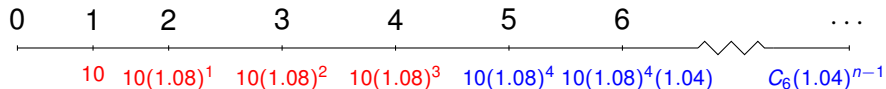
$$PV_0 = 43.83 + 146.41 = 190.25$$

Trick 2

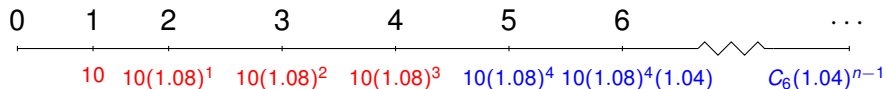
Previously, we used



But what if



Trick 2



Method 2.

$$PV_0 \text{ annuity} = \frac{C}{r-g} \left[1 - \left(\frac{1+g}{1+r} \right)^T \right] = \frac{10}{.02} \left[1 - \left(\frac{1.08}{1.1} \right)^4 \right]$$

$$PV_0 \text{ d,g perpetuity} = \frac{C}{r-g} \left(\frac{1}{1+r} \right)^{t-1} = \frac{10(1.08)^4}{.10 - .04} \left(\frac{1}{1.1} \right)^4$$

$$PV_0 \text{ Total} = 35.38 + 154.87 = 190.25$$

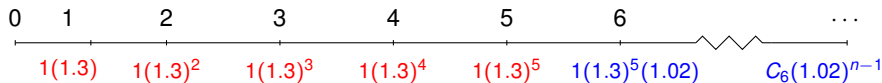
(Slide 37 in FADM Session 1 part 2)

A company just announced earnings of €1. Analysts predict that earnings will grow at 30% per year for the next 5 years. After that, competition is expected to increase, and earnings growth will slow to 2% per year and continue at that level forever. Assume $r = 8\%$. What is the PV of all future earnings?

Example

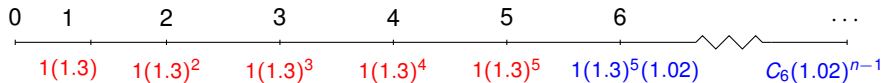
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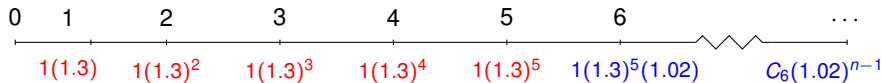
$$PV_0 = \text{PV of growing annuity} + \text{PV of deferred, growing perpetuity}$$

Example



$$\begin{aligned}PV_0 \text{ growing annuity} &= \frac{C}{r - g} \left[1 - \left(\frac{1 + g}{1 + r} \right)^T \right] \\&= \frac{1 \times 1.3}{.08 - .3} \left[1 - \left(\frac{1.3}{1.08} \right)^5 \right] \\&= 9.02\end{aligned}$$

Example



$$\begin{aligned}\text{PV of d,g perpetuity} &= \frac{C}{r-g} \left(\frac{1}{1+r} \right)^{t-1} \\ &= \frac{1 \times (1.3)^5 \times 1.02}{.08 - .02} \left(\frac{1}{1.08} \right)^5 \\ &= 42.96\end{aligned}$$

$$PV_0 \text{ Total} = 9.02 + 42.96 = 51.98$$

